Question			Answer	Marks	AOs		Guidance
12	(a)		$\frac{dy}{dx} = \frac{\frac{x}{x} - \ln x}{x^2}$ oe $\frac{1-\ln x}{x^2} = 0 \Rightarrow \ln x = 1 \Rightarrow \ x = e$	M1 A1 E1 [3]	1.1a 1.1 2.2a	M1 for attempt to use quotient rule (allow one error) Convincing completion (AG)	Subbing x = e gets M0
12	(b)		$\frac{dy}{dx} = \frac{1}{x^2} - \frac{\ln x}{x^2}$ $\frac{d^2y}{dx^2} = -\frac{2}{x^3} - \frac{\frac{x^2}{x} - 2x\ln x}{x^4}$ $= -\frac{2}{x^3} - \frac{x - 2x\ln x}{x^4}$ or $\frac{x^2\left(-\frac{1}{x}\right) - (1-\ln x)2x}{x^4}$ $= \frac{-3 + 2\ln x}{x^3}$ When $x = e$, $\frac{d^2y}{dx^2} = -\frac{1}{e^3} < 0$ hence maximum	M1 A1 E1 [3]	1.1a 1.1 2.1	Attempt to differentiate again (allow one error) Correct second derivative Correct conclusion from correct working	OR M1 subst (must be seen) values either side of e into derivative A1 correct conclusion about sign of gradient E1 correct conclusion from correct working regarding maximum (-0.05)
12	(c)		$\frac{\ln e}{e} > \frac{\ln a}{a} \Rightarrow \frac{1}{e} > \frac{\ln a}{a}$ or $a\ln e > e\ln a$ $e^{\frac{a}{e}} > a$ hence $e^a > a^e$ or $\ln e^a > \ln a^e$ $e^a > a^e$	M1 A1 [2]	3.1a 2.4	Convincing completion (AG)	